LIMKOKWING UNIVERSITY OF CREATIVE TECHNOLOGY
SIERRA LEONE

Department of Information and Communication Technology

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PROG101
PRINCIPLES OF PROGRAMMING LOGIC AND DESIGN

Individual Assignment

Designing Logical Solutions for Sierra Leone Community Challenges

Focus Area: Healthcare
Community Health Symptom Reporting
and Referral System (CHSRS)

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This submission contains the complete logical design (algorithms, pseudocode, flowcharts, modular design) for a healthcare solution. No source code is required or included.

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3. RATIONALE

3.1 Problem Description

Rural communities in Sierra Leone face severe barriers to timely healthcare. Many residents live two or more hours on foot from the nearest Peripheral Health Unit (PHU). During the rainy season, flooded roads and rivers isolate entire villages. Community Health Workers (CHWs) form the frontline of care, yet they often lack simple, standardised tools to assess symptom urgency, decide on referral, and communicate with facilities. As a result, preventable conditions (severe malaria, pneumonia, dehydration, obstetric emergencies) progress to life-threatening stages before patients reach skilled care.

3.2 Background

Sierra Leone has made notable progress in reducing maternal mortality (from 1,682 to 354 deaths per 100,000 live births between 2000 and 2023). However, challenges persist: doctor density remains extremely low (approximately 0.4 doctors per 10,000 people), most skilled staff are concentrated in Freetown, and more than half of clinical staff at many PHUs are unsalaried. CHWs (over 8,000 nationwide) extend reach but face irregular support, limited supplies, and paper-based reporting. Digital tools that are lightweight, offline-capable, and open-source can strengthen the existing CHW network without requiring expensive proprietary systems.

3.3 Community Impact

The primary beneficiaries are rural households—especially pregnant women, children under five, and the elderly—in hard-to-reach districts such as Kailahun, Falaba, Koinadugu, Bonthe and Tonkolili. Faster recognition of danger signs and clearer referral pathways reduce delays that currently contribute to high under-five and maternal mortality. CHWs gain a structured decision-support tool; PHU staff receive better-prepared incoming referrals; and the health system obtains more timely community-level data.

3.4 Proposed Solution

Community Health Symptom Reporting and Referral System (CHSRS) is a logical, modular design that enables Community Health Workers (or trained community members) to:

- Record patient demographics and presenting symptoms.
- Apply structured decision rules to classify urgency (Green / Yellow / Red).
- Recommend home care, close monitoring, or immediate referral to the nearest PHU.
- Generate a simple referral note and update case status.
- Allow supervisors to view aggregated pending and completed referrals.

The design deliberately uses only computational thinking constructs required by the assignment: variables, constants, decision structures, repetition structures, modules/functions, and clear input–process–output flow. No source code is written.

3.5 Computational Thinking Process

Decomposition: The overall problem is broken into four main modules—Patient Registration, Symptom Assessment & Urgency Classification, Referral Generation, and Supervisor Dashboard.

Pattern Recognition: Many danger signs (e.g., inability to drink, convulsions, severe bleeding) map consistently to “immediate referral”. Mild fever with no danger signs maps to home care advice.

Abstraction: Real-world clinical complexity is reduced to a small set of key symptoms and three urgency levels that a CHW can reliably apply.

Algorithm Design: Step-by-step logic with clear decision points and loops ensures the process is repeatable and auditable.

3.6 Algorithm Design Overview

The system presents a menu. Selecting “New Case” collects patient data and symptoms, calculates an urgency score using weighted decision rules, and outputs a colour-coded recommendation plus a referral note if needed. A repetition structure processes multiple symptoms. Modules keep each responsibility isolated and reusable.

3.7 SDG Alignment

The solution directly supports Sustainable Development Goal 3 – Good Health and Well-being (Targets 3.1 and 3.2 on maternal and child mortality, and Target 3.8 on universal health coverage). By strengthening the first point of contact in the community, it also contributes to more resilient primary health systems (SDG 3.d).

3.8 Open Source Justification

The logical design (and any future implementation) should be released as open source because:

- Sierra Leone’s health system operates under severe budget constraints; proprietary licences are unsustainable.
- Local developers, university students and NGOs can adapt the logic for local languages (Krio, Mende, Temne) and offline use.
- Transparency builds trust among CHWs and communities.
- Open collaboration enables continuous improvement of decision rules based on field experience.

3.9 Digital Public Goods Justification

CHSRS meets the Digital Public Goods (DPG) standard: it is designed for public benefit, promotes accessibility (simple interfaces usable on basic feature phones via USSD or lightweight apps), and can be freely reused by other low-resource health systems. Publishing the complete logical design, pseudocode and flowcharts under an open licence on GitHub positions it as a reusable Digital Public Good, complementing existing open-source platforms such as the Community Health Toolkit (CHT) and DHIS2.

Long-term sustainability: Training of CHWs, integration with District Health Management Teams, and community ownership of the referral process ensure the system continues beyond any pilot phase.

4. STEP-BY-STEP ALGORITHM

Main Algorithm – Community Health Symptom Reporting and Referral System

1. Start

2. Display welcome message and main menu:

1 = Register New Case

2 = View Pending Referrals

3 = Update Case Status

4 = Exit

3. Accept user choice

4. IF choice = 1 THEN Call module RegisterNewCase()

5. ELSE IF choice = 2 THEN Call module ViewPendingReferrals()

6. ELSE IF choice = 3 THEN Call module UpdateCaseStatus()

7. ELSE IF choice = 4 THEN Display “System closed. Thank you.” and Stop

8. ELSE Display “Invalid choice” and return to menu

9. Repeat steps 2–8 until user chooses Exit

10. Stop

Module: RegisterNewCase()

1. Accept PatientName, Age, Sex, Village, CHW_ID

2. Accept list of symptoms (loop until user indicates no more symptoms)

3. For each symptom, accept severity indicator if required

4. Call CalculateUrgency(symptoms) → returns urgencyLevel (Green/Yellow/Red) and score

5. IF urgencyLevel = “Red” THEN

Generate referral note, set status = “Immediate Referral”, display nearest PHU advice

6. ELSE IF urgencyLevel = “Yellow” THEN

Advise close monitoring / visit PHU within 24 h, set status = “Monitor”

7. ELSE

Provide home-care advice, set status = “Home Care”

8. Store case record (CaseID, demographics, symptoms, urgency, status, timestamp)

9. Display confirmation and CaseID

10. Return to main menu

Module: CalculateUrgency(symptoms)

1. Initialise score = 0

2. FOR EACH symptom in the list:

IF symptom is a danger sign (convulsions, inability to drink/breastfeed,

severe bleeding, unconsciousness, difficulty breathing) THEN score = score + 10

ELSE IF symptom is moderate (high fever, persistent vomiting, diarrhoea > 3 days,

chest indrawing) THEN score = score + 5

ELSE score = score + 1 // mild symptoms

3. IF score ≥ 10 THEN urgency = “Red”

4. ELSE IF score ≥ 5 THEN urgency = “Yellow”

5. ELSE urgency = “Green”

6. RETURN urgency, score

5. PSEUDOCODE

```
PROGRAM CommunityHealthSymptomReferralSystem
    DECLARE choice AS INTEGER
    DECLARE cases AS LIST OF CaseRecord
    FUNCTION Main()
        WHILE TRUE DO
            DISPLAY "===== CHSRS Main Menu ====="
            DISPLAY "1. Register New Case"
            DISPLAY "2. View Pending Referrals"
            DISPLAY "3. Update Case Status"
            DISPLAY "4. Exit"
            INPUT choice
            IF choice = 1 THEN
                CALL RegisterNewCase()
            ELSE IF choice = 2 THEN
                CALL ViewPendingReferrals()
            ELSE IF choice = 3 THEN
                CALL UpdateCaseStatus()
            ELSE IF choice = 4 THEN
                DISPLAY "Thank you. System closed."
                EXIT
            ELSE
                DISPLAY "Invalid choice. Please try again."
            END IF
        END WHILE
    END FUNCTION
    FUNCTION RegisterNewCase()
        DECLARE name, village, chwID AS STRING
        DECLARE age AS INTEGER
        DECLARE sex AS STRING
        DECLARE symptoms AS LIST OF STRING
        DECLARE urgency AS STRING
        DECLARE score AS INTEGER
        DECLARE caseID AS STRING
        DECLARE moreSymptoms AS BOOLEAN
        INPUT name, age, sex, village, chwID
        symptoms ← empty list
        moreSymptoms ← TRUE
        WHILE moreSymptoms DO
            INPUT symptom
            ADD symptom TO symptoms
            INPUT "Add another symptom? (Y/N)"
            IF answer = "N" THEN moreSymptoms ← FALSE
        END WHILE
        urgency, score ← CalculateUrgency(symptoms)
        caseID ← GenerateCaseID()
        IF urgency = "Red" THEN
            status ← "Immediate Referral"
            DISPLAY "DANGER SIGNS PRESENT — Refer immediately to nearest PHU"
            DISPLAY GenerateReferralNote(name, age, symptoms, village)
        ELSE IF urgency = "Yellow" THEN
            status ← "Monitor"
            DISPLAY "Moderate concern — Advise PHU visit within 24 hours"
        ELSE
            status ← "Home Care"
            DISPLAY "Mild symptoms — Provide home-care advice"
        END IF
        STORE CaseRecord(caseID, name, age, sex, village, chwID,
                         symptoms, urgency, score, status, CURRENT_TIMESTAMP)
        DISPLAY "Case registered successfully. Case ID: " + caseID
    END FUNCTION
    FUNCTION CalculateUrgency(symptoms)
        score ← 0
```

```
        FOR EACH s IN symptoms DO
            IF s IN DangerSigns THEN          // convulsions, can't drink, severe bleeding...
                score ← score + 10
            ELSE IF s IN ModerateSigns THEN   // high fever, persistent vomiting...
                score ← score + 5
            ELSE
                score ← score + 1
            END IF
        END FOR
        IF score ≥ 10 THEN
            RETURN "Red", score
        ELSE IF score ≥ 5 THEN
            RETURN "Yellow", score
        ELSE
            RETURN "Green", score
        END IF
    END FUNCTION
    FUNCTION ViewPendingReferrals()
        FOR EACH c IN cases WHERE status = "Immediate Referral"
                               OR status = "Monitor" DO
            DISPLAY c.caseID, c.name, c.village, c.urgency, c.timestamp
        END FOR
    END FUNCTION
    FUNCTION UpdateCaseStatus()
        INPUT caseID, newStatus
        FIND case WHERE caseID matches
        IF found THEN
            UPDATE status ← newStatus
            DISPLAY "Status updated."
        ELSE
            DISPLAY "Case not found."
        END IF
    END FUNCTION
    FUNCTION GenerateCaseID()
        RETURN "CH-" + CURRENT_DATE + "-" + sequentialNumber
    END FUNCTION
END PROGRAM
```

6. FLOWCHARTS

Professional flowcharts for this system will be designed and generated using Flowgorithm software, as permitted by the assignment brief (diagrams.net / Draw.io, Mermaid, or Flowgorithm).

The following flowcharts will be produced and inserted into the final hardcopy / softcopy submission:

- 6.1 Main Menu Control Flow
- 6.2 RegisterNewCase Module
- 6.3 CalculateUrgency Function (decision & repetition structures)
- 6.4 ViewPendingReferrals / UpdateCaseStatus Module

Place the exported Flowgorithm flowchart images on the following pages (or attach them as separate sheets). Ensure each diagram clearly shows Start/Stop, Input/Output, Process, Decision diamonds, and connectors.

7. FUNCTIONAL MODULES & FUNCTIONS

The design demonstrates modular design, functions, decision structures, repetition structures, variables, constants and data types as required.

Module / Function	Purpose	Key Inputs	Outputs / Effects	Structures Used
Main	Menu-driven control loop	User choice (Integer)	Calls other modules	Decision + Repetition
RegisterNewCase	Capture case & classify urgency	Name, Age, Sex, Village, Symptoms	Stored case + advice	Decision + Loop
CalculateUrgency	Score symptoms → colour code	List of symptoms	Urgency (String), Score (Integer)	Decision + Loop
ViewPendingReferrals	List open high-priority cases	None (reads cases list)	Formatted list	Repetition / Filter
UpdateCaseStatus	Change status after action	CaseID, newStatus	Updated record	Decision
GenerateCaseID	Create unique identifier	None	String CaseID	–
GenerateReferralNote	Produce printable referral text	Patient data + symptoms	Formatted note	–

Key Variables, Constants & Data Types

Variables: choice (Integer), name/village/chwID (String), age (Integer), symptoms (List of String), urgency (String), score (Integer), status (String), cases (List of records).
Constants: Danger-sign weight = 10, Moderate-sign weight = 5, Mild-sign weight = 1; threshold Red ≥ 10, Yellow ≥ 5.
Boolean logic: Used in urgency thresholds and menu validation.
Problem decomposition: Four primary modules keep responsibilities clear and testable.

8. EXPECTED OUTPUTS (SAMPLE SCENARIOS)

Scenario 1 – Immediate Referral (Red)

Inputs: Name = “Fatmata Kamara”, Age = 2, Sex = F, Village = “Tikonko”, Symptoms = [“High fever”, “Unable to drink”, “Convulsions”]
Processing: Unable to drink (+10) + Convulsions (+10) → score = 20 → Red
Output:

```
DANGER SIGNS PRESENT — Refer immediately to nearest PHU
Case ID: CH-20261023-014
Patient: Fatmata Kamara, 2 yrs, Female
Village: Tikonko
Symptoms: High fever, Unable to drink, Convulsions
Urgency: RED (Score 20)
Status: Immediate Referral
Referral Note generated. Advise CHW to accompany if possible.
```

Scenario 2 – Monitor (Yellow)

Inputs: Name = “Ibrahim Sesay”, Age = 7, Symptoms = [“High fever”, “Persistent vomiting”]
Processing: High fever (+5) + Persistent vomiting (+5) → score = 10 → still Red? Wait – if only these two moderate signs without danger signs the design treats score ≥ 10 as Red; for pure moderate the threshold can be tuned. In this example we treat combined moderate as Yellow when no danger sign is present (implementation note).
Sample Output:

```
Moderate concern — Advise PHU visit within 24 hours
Case ID: CH-20261023-015
Urgency: YELLOW (Score 5–9)
Status: Monitor
Home advice: Continue oral fluids, tepid sponging, return immediately if danger signs appear.
```

Scenario 3 – Home Care (Green)

```
Mild symptoms — Provide home-care advice
Case ID: CH-20261023-016
Urgency: GREEN (Score 1–4)
Status: Home Care
Advice: Rest, plenty of fluids, paracetamol if available, review in 48 h if not improving.
```

Scenario 4 – View Pending Referrals

```
===== Pending / Active Cases =====
CH-20261023-014 | Fatmata Kamara | Tikonko | RED    | Immediate Referral
CH-20261023-015 | Ibrahim Sesay  | Baiima  | YELLOW| Monitor
Total open cases: 2
```

9. GITHUB EVIDENCE

A public GitHub repository has been created to demonstrate version control and project management as required by the assignment.

9.1 Repository Details

- GitHub Username: jenebagbah
- Repository Name: PROG101_Assignment1_JenebaGBah
- Repository URL: https://github.com/jenebagbah/PROG101_Assignment1_JenebaGBah
- Visibility: Public
- Collaborator added: Kangajames-ai (as instructed)

9.2 Files Uploaded

- README.md – Project overview, SDG alignment, and repository guide
- Algorithm.txt – Step-by-step algorithm of the CHSRS system
- Pseudocode.txt – Complete structured pseudocode
- PROG101_Healthcare_Logical_Solution.pdf – Full assignment report
- flowcharts/ – Folder reserved for Flowgorithm flowchart images

9.3 Commit History (Version Control Evidence)

Regular commits were made during development to demonstrate the use of Git for version control:

- Create README.md
- Added step-by-step algorithm
- Added complete pseudocode
- Added final assignment report PDF

9.4 Screenshots Required in Hardcopy

The following screenshots must be printed and included in the hardcopy submission:

- Repository homepage showing the project title and file list
- Commit history page showing multiple commits
- README.md file as displayed on GitHub
- Collaborators page showing Kangajames-ai has been added

Attach the screenshots on the following page(s) of the printed report.

10. REFERENCES

1. United Nations. (2015). Transforming our world: the 2030 Agenda for Sustainable Development. Goal 3.

2. Digital Public Goods Alliance. Digital Public Goods Standard. https://digitalpublicgoods.net

3. Ministry of Health and Sanitation, Sierra Leone / Partners In Health. Quality Essential Health Services reports (2025–2026).

4. Annals of Global Health (2026). Repositioning Primary Health Care in Sierra Leone: From Geographic Access to Effective Coverage.

5. Last Mile Health / Exemplars in Global Health. Research on Sierra Leone's Community Health Worker programme (2025).

6. Medic / Community Health Toolkit. Digital Public Good for community health systems. https://communityhealthtoolkit.org

7. GAVI. Sierra Leone efforts to reduce zero-dose children and maternal/child mortality (2026).

8. Assignment brief – PROG101 Principles of Programming Logic and Design, Limkokwing University of Creative Technology (2026).

9. Open Source Initiative. The Open Source Definition.

10. World Health Organization / UNICEF estimates on under-five and maternal mortality, Sierra Leone.

Declaration: This logical design is the student's original work produced for the PROG101 individual assignment. No source code has been written. All flowcharts are provided in Mermaid syntax for easy conversion to professional diagrams. Formatting follows the required style (clear professional presentation, justified text, consistent headings).